

Retrotransposon self-insertion in HG03086

Question (W. Brandler): do MEI insertions favour inserting into existing reference copies of their own family, and when they do, is the new insertion in the same orientation as the reference element? Following his clarification — *"we would want to annotate nested_sense and nested_antisense and compare the two"* — the two orientation classes are treated as separate categories throughout rather than as a single same-orientation rate.

Answers. Same-family nesting exceeds the selected placement nulls for Alu and LINE-1. For SVA the result is 9 observed vs 0.306 expected (n=9; repeat-aware callability untested), so it remains provisional. Nummi et al. 2025 report SVA self-target preference, providing systematic precedent but not validation of these nine calls. The nested set has a strong same-orientation tendency for Alu and LINE-1; within-nested tests and category-vs-placement tests are distinct. Among AluY insertions, young-host targeting is above an abundance-based expectation (1.86×), an age-associated pattern compatible with—but not proof of—homology- mediated targeting. The orientation tendency remains unexplained by these analyses: tested motif/opportunity models do not explain its magnitude, and no enrichment was detected for CCATT under the selected control. These results do not assign or exclude a route.

Scope note. This is the table-driven report and retains exploratory subfamily and motif sections. See the integrated evidence report (NESTED_MEI_INTEGRATED_REPORT_2026-10-02.md) for the current literature-reconciled synthesis and the validation input guide (SELFINS_VALIDATION_INPUT_GUIDE.md) for the blocked sequence/read-backed checks. The figures, tables, and quantitative summaries below are rendered into both this Markdown and PDF by this script.

Callset: 1852 classifier calls (score ≥ 0.997) on HG03086, GRCh38, chr1–22 + chrX — one genome, which bounds several results below. Reference elements: UCSC hg38 RepeatMasker, filtered to Alu / LINE-1 / SVA with the pipeline's own family-normalisation rule (`<font name="Courier">_normalize_mei_family_token</font>`). 320 of 1852 calls (17.3%) fall inside a same-family element.

1. Same-family nesting against family-specific selected nulls

Two nulls, each resampling call positions genome-wide 1,000 times — **uniform**, and **GC-matched** on each call's local GC (1 kb window). Empirical p-values at 0.000999 are at the simulation floor, not precise tail-probability estimates.

family	n	nested	observed	GC-matched exp.	uniform exp.	conservative enrichment	p
Alu	1555	239	0.154	0.096	0.105	1.47×	< 0.001
LINE-1	222	72	0.324	0.220	0.177	1.47×	< 0.001
SVA	75	9	0.120	0.004	0.001	29.41×	< 0.001

The selected null differs by family. GC matching yields an Alu expectation of 149.6 vs 162.5 under uniform (1.60× vs 1.47×), and raises expectations for LINE-1 and SVA relative to uniform. These are model-dependent placement comparisons; neither null is proven universally correct. Polymorphic AluY in this callset have lower mean GC than fixed reference AluY (0.385 vs 0.441). This descriptive contrast does not by itself separate integration preference from sequence composition, ascertainment, or survival/selection.

2. Sense and antisense, as separate categories

Host-element selection is **orientation-blind**: where several same-family elements overlap a breakpoint the longest is chosen, never the one whose strand matches. The pipeline's own `<font name="Courier">_annotate_nested_retrotransposon</font>` scores candidates `<font name="Courier">(same_orient, length)</font>`, so reusing it would have folded the tie-break into the measurement.

family	nested	same orientation	rate	95% CI	odds ratio	Fisher p
Alu	239	202	0.845	0.793–0.889	30.0	1.4e-28
LINE-1	72	53	0.736	0.619–0.833	7.2	0.000202
SVA	9	9	1.000	0.664–1.000	∞	0.00794

Splitting the two classes and testing each against the null separately is where the single rate turns out to be hiding something:

family	category	observed	expected	enrichment	p
Alu	nested_sense	202	81.27	2.49×	3.4e-29
Alu	nested_antisense	37	81.27	0.46×	6.2e-08
LINE-1	nested_sense	53	24.43	2.17×	7.6e-07
LINE-1	nested_antisense	19	24.43	0.78×	0.318
SVA	nested_sense	9	0.15	58.82×	2.2e-13
SVA	nested_antisense	0	0.15	0.00×	1

Against the orientation-blind placement simulations, Alu antisense nesting is below the selected null (0.46×, p = 6.2e-08), whereas LINE-1 antisense is compatible with its null (0.78×, p = 0.318). These are category-vs-placement tests, not a direct sense-vs-antisense contrast among nested calls. The conditional same-orientation fractions and their binomial tests are reported separately above; Alu's marginal-based expectation (0.500) is close to, but not conceptually identical with, the 0.5 binomial baseline.

The chance baseline is close to the marginal-based expectation, but the tests are distinct. For Alu, new insertions are 0.502 + and reference hosts are 0.515 +, giving an independence-expected same-orientation rate of 0.500. The saved conditional exact-binomial test uses 0.5; the saved Fisher test uses the actual orientation marginals. These are similar here, not the same model in general.

A limited subfamily negative control

Nested calls carry different subfamily-age distributions from non-nested calls, which argues against one simple host-sequence misassignment explanation. It is not a general exclusion of calling or mapping artefacts; the orientation split by age is a robustness check, not read-level validation:

family	subset	n	same orientation	p vs 0.5
Alu	young only	140	0.843	4.5e-17
Alu	old only	99	0.848	8.2e-13
LINE-1	young only	36	0.694	0.0288
LINE-1	old only	36	0.778	0.00119

Only 5.9% of nested Alu calls carry a subfamily label identical to their host element (0.0% for LINE-1). This argues against one simple same-label host re-detection explanation; it is not a general exclusion of calling or mapping artefacts and does not independently validate the calls.

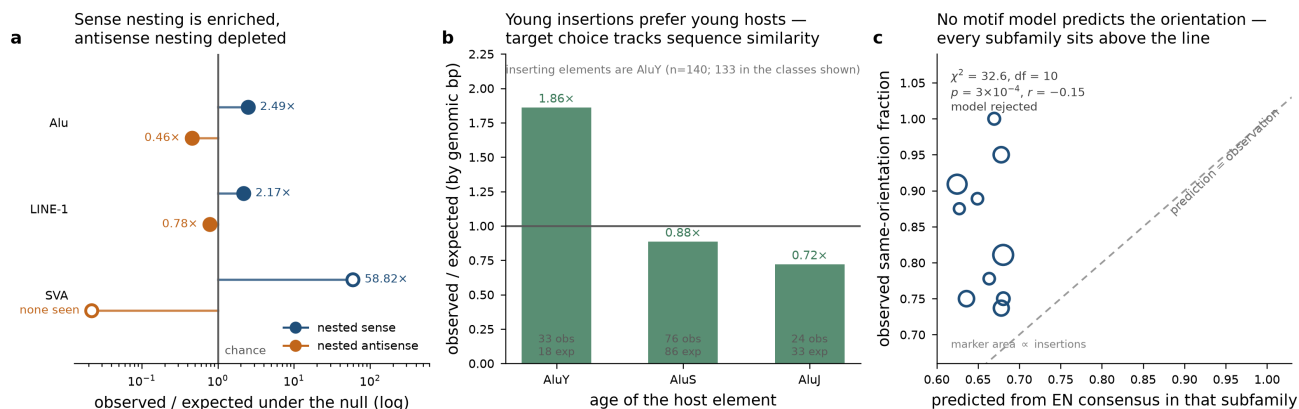


Figure 1. Self-insertion in HG03086: (a) sense and antisense nesting tested separately against the null; log scale, open markers mark SVA as underpowered. (b) where young AluY insertions land by host-element age, expected in proportion to each class's genomic bp. (c) observed same-orientation fraction against the degenerate EN consensus, per host subfamily; marker area scales with insertion count.

3. AluY host-age pattern (sequence similarity is not isolated)

Restricting to young AluY-subfamily insertions and asking which host they land in, against an expectation proportional to each host class's share of genomic Alu bp:

host element	genomic bp	observed	expected	enrichment	p
AluY (youngest)	38.6 Mb	33	17.7	1.86×	0.000762
AluS (intermediate)	187.0 Mb	76	85.9	0.88×	0.154
AluJ (oldest)	72.6 Mb	24	33.3	0.72×	0.0573
other	6.5 Mb	7	3.0	2.35×	0.0326

The final row is a heterogeneous remainder (FLAM, FRAM and other non-age-graded Alu-family annotations); it is listed so the table accounts for all 140 insertions rather than only the 133 that fall on the age axis, and it carries no interpretation.

The abundance-based table shows an age-associated distribution: young AluY insertions favour young hosts (1.86×, $p = 0.000762$), while the oldest AluJ class is below expectation (0.72×, $p = 0.0573$). Host age is only a proxy for sequence similarity and also covaries with composition, genomic location, and survival; this result does not by itself demonstrate homology-directed integration. Jacob-Hirsch et al. 2018 describe homology/abundance effects for a distinct somatic-brain LINE-1 class. The direction is qualitatively compatible, but the cohorts and mechanisms differ, so this is not a direct replication.

This age-associated pattern does not explain orientation: same-orientation fractions are similar across host-age strata (0.875 AluJ, 0.829 AluS, 0.818 AluY, $n = 24 / 76 / 33$). A simple model in which host homology drives the orientation preference would predict a stronger same-orientation fraction in more similar hosts; these data do not show that trend.

Where inside the host element insertions land: two A-rich peaks

Levy, Schwartz & Ast (2009 online; 2010 issue, *Nucleic Acids Research* 38:1515–1530, doi:10.1093/nar/gkp1134; Results p. 1518, Figure 2) report a prominent Alu-into-Alu insertion hotspot directly after the A-rich linker at targeted-Alu consensus coordinate 133. Their Figure 2 describes the AluJo consensus and labels the internal A-rich linker at 118–136; Figure 1 defines insertion position as the coordinate just upstream of the insertion. Under a 5'-aligned consensus-to-host convention, coordinate 133 falls in our 120–140 bp bin, with the breakpoint around the 133/134 boundary—not an exact bin edge. The paper does not explicitly say that 133 is counted from the first consensus nucleotide, so that mapping is an inference. Figure 2's sense-oriented Alu self-insertion display uses 15,759 Alu self-insertion events. Our 20-bp profile is positionally compatible, not a nucleotide-resolution replication.

The HG03086 distribution across deciles of the host element is strongly non-uniform — $\chi^2 = 103$ on 9 df, $p = 3.7e-18$, with a Kolmogorov–Smirnov test against uniform giving $D = 0.161$, $p = 7.0e-06$ ($n = 239$ nested Alu calls).

Restricting to near-full-length host Alus (154 calls, host 280–320 bp) so base offsets are comparable, and binning at 20 bp against a uniform expectation of 9.6 per bin:

position in host	insertions	host A-fraction	region
280–300 bp	36 (3.7×)	0.79	3' poly-A tail
120–140 bp	29 (3.0×)	0.55	inter-arm A-rich linker
0–20 bp	13 (1.4×)	0.13	5' end — not A-rich

The separate 280–300 bp peak (36, 3.7×) is near the terminal poly(A) tail under this host convention. Levy et al. excluded target-element consensus ends from positional tests because of RepeatMasker/short-flank identification limitations, so their data do not directly adjudicate our terminal-tail cluster. This is a coverage blind spot, not evidence by itself that the cluster is novel or genuine. Nummi et al. 2025 independently report germline Alu target peaks at 100–155 and 260–310 bp, broadly covering both of our peak windows, but their long-read cohort and positional method differ.

Insertion count correlates with host A-fraction across bins (Pearson $r = +0.513$, $p = 0.0422$) but not rank-wise (Spearman $\rho = +0.063$, $p = 0.815$): the association is carried by discrete peaks, not a smooth gradient. Neither positional peak validates its individual insertions as authentic TPRT events.

This bears on §4. The motif models tested there fail to predict the orientation *magnitude*. A-rich target positions coincide with the two larger positional peaks, but this positional association does not explain the orientation result.

The 5'-most bin carries 13 insertions at low A-content (0.13) and has no mechanistic account. It sits at an element boundary, where breakpoint-assignment edge effects are the leading explanation; it is flagged rather than interpreted.

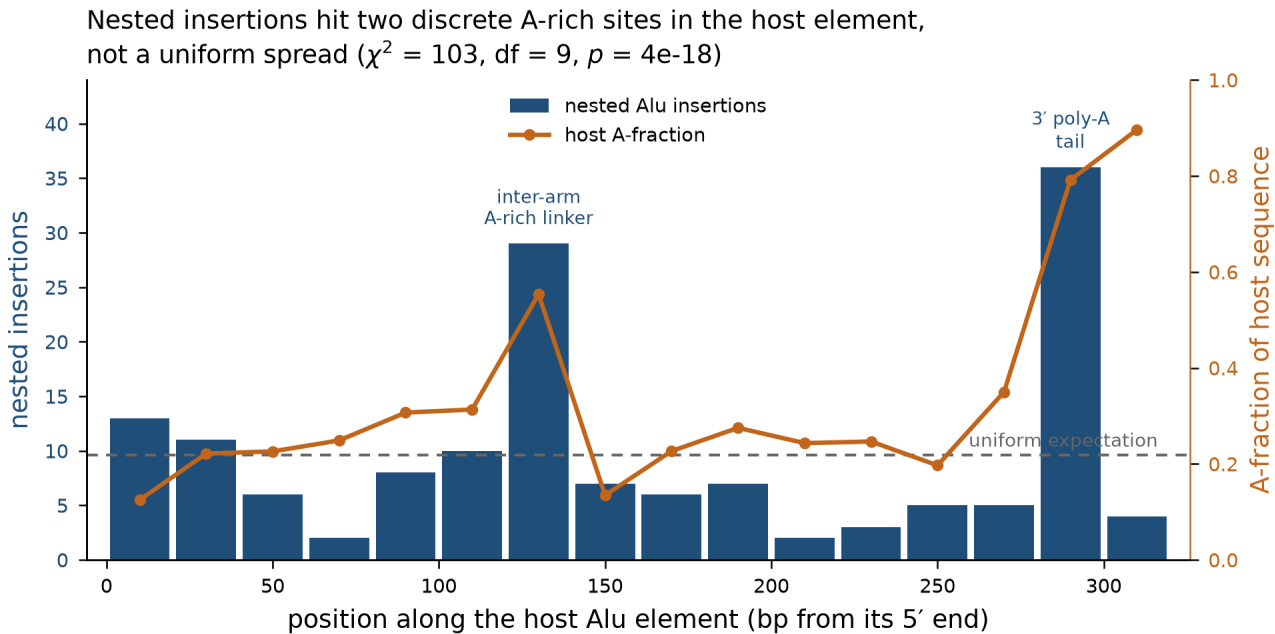


Figure 2. Nested Alu insertion positions within the host element. Bars show insertion counts in 20 bp bins for near-full-length hosts; line shows mean host A-fraction in element orientation; dashed line is the uniform expectation.

4. Orientation: tested models do not explain the observed tendency

The age-associated host-choice result in §3 does not account for orientation. Three sequence/opportunity hypotheses are evaluated below; none explains the observed magnitude.

Narrow endonuclease consensus (TTTT/AA). Counting sense- and antisense-compatible sites inside host elements predicted the family-level rate closely — Alu 0.842 against observed 0.845, LINE-1 0.731 against 0.736. That agreement does not survive. It was computed over elements weighted by *genomic abundance*, dominated by old low-prediction copies; weighted by the subfamilies that actually received insertions the same model predicts 0.920 against an observed 0.842. Two weightings landing near each other is not a prediction, and across subfamilies the model was rejected outright.

Full degenerate consensus (YY/RRRR). `TTTT/AA` is one instance of the published L1 EN consensus; scoring only it misses every other variant. Recounting on the full degenerate pattern:

family	predicted (degenerate)	predicted (narrow)	observed
Alu	0.658	0.842	0.845
LINE-1	0.647	0.731	0.736
SVA	0.679	0.834	1.000

The tested full degenerate consensus predicts a lower same-orientation fraction than observed for each family. Across host subfamilies its predictions have little range relative to the observed fractions. This model does not account for the observed magnitude; it does not establish that endonuclease opportunity is biologically irrelevant.

Method check: genome-wide the two consensus orientations occur at 203.3 and 201.4 sites per 10 kb — ratio **1.009**. This background check finds no global strand skew in the motif counts; it does not by itself explain the nested-call orientation pattern.

CCATT target motif (literature context, not route assignment). Jacob-Hirsch et al. report a CCATT-associated EN-independent route for a specific somatic-brain nested L1 class. Against a control matched for element composition—random positions *inside same-family elements*, since nested sites sit in element sequence by construction—no enrichment was detected in this callset under this control: 2.06 vs 1.79 per kb, rate ratio 1.15, **p = 0.445**. This does not establish that these calls used the classical route; route assignment needs independent junction, truncation, and target-motif evidence.

The observed orientation tendency is pronounced in these nested calls, but remains unexplained by the tested sequence/opportunity models. These tests do not establish that the corresponding biological mechanisms are absent.

Levy et al.'s “~4-fold” figure has a different unit than this report's enrichment. The “~4-fold” figure from Levy et al. 2010 is not the ratio of total insertions into TE sequence to insertions into non-TE sequence. It is the ratio of enriched to depleted insertion-type categories: 516 overrepresented versus 116 underrepresented insertion-type categories out of 36,478 tested (young × old × orientation), i.e. ~4.45:1 by category. Their per-type null is random insertion into unique intergenic DNA scaled to each old TE class's inferred ancestral coverage. Any total-event ratio must be computed from the underlying event counts, which the paper does not report as a single number.

That category-count result is not directly comparable to §1's family-specific observed-vs-resampled event counts; the two answer different statistical questions. Levy et al.'s exact-insertion database is a broader set than the young-into-old unique-intergenic analysis.

These tests identify hypotheses not supported by the current measurements; they do not rule out the corresponding biological mechanisms. The orientation tendency remains unexplained by this analysis.

5. Pipeline annotation context (separate from validation of this callset)

Earlier pipeline output collapsed same-family opposite-orientation overlaps into the `unnested` label, so those records could not be separated from genuinely non-nested calls using that field alone. In the analysis-table reconstruction, 56 same-family opposite-orientation calls were identified (37 Alu, 19 LINE-1). This describes a pipeline-annotation limitation and is not read-level validation of the current callset.

The analysis reconstructed the two orientation classes independently of that collapsed field. A separate pipeline branch documented in the prior project handoff added a four-valued `NESTED` enum (branch/test details are not independently reverified in this report):

value	meaning
<code>unnested</code>	no same-family element overlaps the breakpoint

value	meaning
nested_sense	overlaps; insertion orientation matches the element strand
nested_antisense	overlaps; orientations differ
nested_unknown	overlaps, but orientation is unresolvable on one side

nested_unknown retains records whose host or insertion orientation cannot be resolved. For this report's analysis, host selection was reconstructed without preferring matching strands; raw reads were not independently validated.

6. Limitations

- **One genome.** HG03086 is the only genome-wide callset in the project; other samples are chr22 slices. These are catalogue-pattern estimates for one individual, not population-wide integration rates. - **SVA is provisional.** 9 nested calls (9 observed vs 0.306 expected under the selected GC-matched null). Repeat-aware callability/mappability was not modeled, so the 29.4× ratio is not a settled effect estimate. Nummi et al. 2025 report systematic SVA self-target preference, which is precedent—not validation of these nine calls. - **Genic vs intergenic: not assessed.** The sense fraction is 0.851 genic against 0.811 intergenic, but with 37 intergenic events the comparison can only detect a 21-point difference at 80% power. This is absence of power, not absence of effect, and is reported as such. - **Orientation mechanism is open.** The age-associated host-choice result does not explain orientation. Tested motif/opportunity models do not account for its magnitude; this does not prove the corresponding biological mechanisms absent. - **Read-backed validation is incomplete.** Repetitive loci challenge short-read evidence. The current checkout lacks HG03086 BAM/CRAM and the full reference sequence needed to reconstruct junctions; known-source and read-support annotations do not substitute for independent per-locus validation. - **Element boundaries affect the rate, not the orientation result.** An earlier draft asserted that RepeatMasker annotates Alu poly-A tails separately from element bodies, so that tail-primed insertions would fall outside the interval and understate nesting. That was checked and is **false**: on chr1, 0.8% of 98,824 Alu elements have a separately annotated adjacent poly-A/T, so the tail is almost always inside the annotation. The real boundary question was tested directly by relaxing every interval by a fixed slop and recomputing both the nesting flag and the host strand from scratch at each step:

slop	Alu nested	Alu same-orient	LINE-1 nested	LINE-1 same-orient
±0 bp	0.154	0.845	0.324	0.736
±10 bp	0.170	0.848	0.369	0.768
±25 bp	0.188	0.836	0.383	0.765
±50 bp	0.208	0.815	0.423	0.723
±100 bp	0.250	0.787	0.441	0.714
±250 bp	0.329	0.691	0.491	0.679

The nesting **rate** is boundary-dependent, as any interval-overlap statistic is — but the enrichment in §1 compares observed against a null built from the same intervals, so the comparison is like-for-like at whatever boundary is used. The **orientation** result is robust: still 0.815 for Alu and 0.723 for LINE-1 at ±50 bp, decaying only as flanking non-element sequence — where no orientation relationship exists — is pulled in.

Prior work

year	citation	bearing on this analysis
2009/2010	Levy, Schwartz & Ast	Primary genome-wide nested-TE study. Results p. 1518 and Figure 2 place an Alu-into-Alu hotspot just after the A-rich linker at AluJo consensus coordinate 133 (linker 118–136); under 5' consensus-to-host alignment this falls in our 120–140 bp bin. Figure 1 defines insertion position; Figure 2's sense-oriented Alu self-insertion display uses 15,759 Alu self-insertion events. The breakpoint is near 133/134; mapping is an inference, not a bin assignment in the paper. Target-end exclusion does not test our 280–300 bp tail cluster. Their '~4-fold' figure is 516 overrepresented versus 116 underrepresented insertion-type categories out of 36,478 tested, not an aggregate event ratio. Their per-type null is random insertion into unique intergenic DNA scaled to each old TE class's inferred ancestral coverage; the paper does not report a total-event ratio as a single number.
2010	Kojima 2010	Discusses differences in cis- and trans-mobilized insertion structure. These observations are context for family differences, not a direct prediction of the nested-orientation split measured here.
2018	Jacob-Hirsch et al. 2018	This is mechanistic context for a particular somatic-brain L1 class, not a numerical comparator for inherited mixed-family events. The current CCATT control detects no enrichment under its chosen background; it does not assign the route used by these calls.
2019	Flasch et al. 2019	Mechanistic context for possible alternatives; it does not predict the particular nested orientation counts measured here.
2025	Nummi et al. 2025	Independent systematic precedent for SVA self-target preference and similar broad Alu positional regions. Cohort, ascertainment, and methods differ; these results do not validate individual HG03086 calls or remove its callability caveat.

Reproducing

Run `python artifacts/build_selfins_report.py` from the repository root to regenerate this Markdown and `selfins_report.pdf` from the same report text, tables, and figures. Quantitative summaries are read from saved results; literature context and interpretation are curated in the generator template. PDF rendering requires ReportLab (`python -m pip install -r artifacts/requirements-pdf.txt`).

file	contents
<code>selfins_per_call.csv</code>	all calls with host element, nesting, orientation
<code>selfins_enrichment.csv</code>	both nulls, all three families
<code>selfins_sense_antisense.csv</code>	the two orientation classes tested separately
<code>selfins_orientation.csv</code>	orientation tests with marginals and Fisher results
<code>selfins_orientation_sensitivity.csv</code>	young/old subfamily split
<code>selfins_homology_test.csv</code>	host-age preference for young insertions
<code>selfins_en_degenerate_family.csv</code>	degenerate <code>YY/RRRR</code> recount, family level
<code>selfins_en_degenerate_subfamily.csv</code>	the same per subfamily, where it is rejected
<code>selfins_en_motif_prediction.csv</code>	the earlier narrow <code>TTTT/AA</code> count
<code>selfins_ccatt_control.csv</code>	CCATT control, composition-matched
<code>selfins_genome_background_en.json</code>	genome-wide consensus density, both orientations

file	contents
selfins_position_profile.csv	insertion counts and host A-fraction per 20 bp bin (§3)
selfins_position_deciles.csv	decile counts underlying the uniformity test (§3)
selfins_position_stats.json	χ^2 , KS and correlation statistics for §3
selfins_boundary_sensitivity.csv	nesting and orientation as element boundaries are relaxed
selfins_polya_annotation_check.json	whether rmsk annotates Alu poly-A tails separately
alu_insertion_vs_retention_gc.csv	insertion vs retention GC contrast
nested_orientation_literature.csv	prior work with per-paper bearing